

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows:3,900
- Columns:18
- Key Features:
 - Customer Demographics(Age, Gender, Location, Subscription Status)
 - Purchase Details(Items Purchased, Category, Purchase Amount, Season, Size, Colour)
 - Shopping Behavior(Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping type)
 - Missing Data: 37 values in Review Rating Column

3. Exploratory Data Analysis Using Python

I began with data presentation and cleaning in Python:

- ❖ **Data Loading:** Imported the dataset using `pandas`.
- ❖ **Initial Exploration:** Used `df.info()` to check structure and `df.describe()` to check summary statistics.

	customer_id	age	purchase_amount	review_rating	previous_purchases	purchase_frequency_days
count	3900.000000	3900.000000	3900.000000	3900.000000	3900.000000	3337.000000
mean	1950.500000	44.068462	59.764359	3.750051	25.351538	88.986815
std	1125.977353	15.207589	23.685392	0.713590	14.447125	128.690263
min	1.000000	18.000000	20.000000	2.500000	1.000000	7.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000	14.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000	30.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000	90.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000	365.000000

- ❖ **Missing Data Handling:** Checked for null values and imputed missing values in the **Review Rating** column using the median rating of each product category.
- ❖ **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- ❖ **Feature Engineering:**
 - ❖ Created **age_group** column by binning customer ages.
 - ❖ Created **purchase_frequency_days** column from purchase data.
- ❖ **Data Consistency Check:** Verified if **discount_applied** and **promo_code_used** were redundant: dropped **promo_code_used**.
- ❖ **Database Integration:** Connected Python script to PostgreSQL and loaded the clean DataFrame into the database for SQL analysis

4. Data Analysis Using SQL (Business Transactions)

I performed structure data analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender:** Compared total revenue generated from male vs. female customers.

	gender text 🔒	revenue numeric 🔒
1	Female	75191
2	Male	157890

2. **High-spending discount users:** Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint 🔒	purchase_amount bigint 🔒
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
Total rows: 839		Query complete 00:0

3. **Top 5 products by Rating:** Found products with the highest average review ratings.

	item_purchased 	average_product_rating 
	text	numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Comparison:** Compared average purchase amounts between Standard and Express Shipping.

	shipping_type 	avg 
	text	numeric
1	Standard	58.4602446483180428
2	Express	60.4752321981424149

5. **Subscribers vs. Non-subscribers:** Compared average spending and total revenue across subscription status.

	subscription_status 	total_customers 	average_spending 	total_revenue 
	text	bigint	numeric	numeric
1	Yes	1053	59.4919278252611586	62645
2	No	2847	59.8651211801896733	170436

6. **Discount-dependent products:** Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

7. **Customer Segmentation:** classified customers into New, Returning and Loyal segments based on purchase history.

	customer_segment text	number_of_customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. **Top 3 products per category:** Listed the most purchased products within each category.

	item_rank bigint 🔒	category text 🔒	item_purchased text 🔒	total_orders bigint 🔒
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers and Subscriptions:** Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status text 🔒	repeat_buyers bigint 🔒
1	No	2518
2	Yes	958

10. **Revenue by Age group:** Calculated total revenue contribution of each age group.

	age_group text	revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

Finally, I built an interactive dashboard in Power BI to present insights visually:



6. Business Recommendations:

❖ **Boost Subscriptions:** Promote exclusive benefits for subscribers.

- ❖ **Customer Loyalty Programs:** Reward repeat buyers to move them into the “Loyal” segment.
- ❖ **Review Discount Policy:** Balance sales boosts with margin control.
- ❖ **Product Positioning:** Highlight top-rating and best-selling products in campaigns.
- ❖ **Targeted Marketing:** Focus efforts on high-revenue age groups and express-shipping users.